A Simple SNAKE & LADDER Game Implementation using PYTHON

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import random

# Global variables
dice = 0
key = ''
choice = ''
position = 0
newposition = 0
count = 0

def check_position(position):
  global newposition
  if position == 4:
    newposition = 14
    print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
  elif position == 9:
    newposition = 31
    print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
  elif position == 17:
    newposition = 7
    print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
  elif position == 21:
    newposition = 42
    print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
  elif position == 46:
    newposition = 13
    print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
  elif position == 28:
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        newposition = 84
        print(f"\nWow! You landed on a big ladder. Now you are on {newposition}.")
    elif position == 62:
        newposition = 19
        print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
    elif position == 37:
        newposition = 61
        print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
    elif position == 54:
        newposition = 25
        print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
    elif position == 51:
        newposition = 67
        print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
    elif position == 87:
        newposition = 36
        print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
    elif position == 72:
        newposition = 91
        print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
    elif position == 64:
        newposition = 44
        print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
    elif position == 80:
        newposition = 23
        print(f"\nOh! You landed on a big snake. Now you are on {newposition}.")
    elif position == 38:
        newposition = 71
        print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
    elif position == 95:
        newposition = 75
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        print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
    elif position == 98:
        newposition = 79
        print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
    else:
        newposition = position

    return newposition

def main():
    global dice, position, newposition, count, key, choice

    # Initial greetings
    print("Would you like to Play the game???(click Y or y if you want)")
    choice = input().strip().lower()

    if choice == 'y' or choice=='Y':
        print(" ### Welcome to Snake and Ladders Game !!!!!       | Rules Are::                         |")
        print("|  100  99  98  97  96  95  94  93  92  91  |       | 1=START              4=Ladder to 14 |")
        print("|  81   82  83  84  85  86  87  88  89  90  |      | 9=Ladder to 31         17=Snake to 7  |")
        print("|  80   79  78  77  76  75  74  73  72  71  |      | 21=Ladder to 42        46=Snake to 13 |")
        print("|  61   62  63  64  65  66  67  68  69  70  |      | 28=Ladder to 84        62=Snake to 19 |")
        print("|  60   59  58  57  56  55  54  53  52  51  |      | 37=Ladder to 61        54=Snake to 25 |")
        print("|  41   42  43  44  45  46  47  48  49  50  |      | 51=Ladder to 67        87=Snake to 36 |")
        print("|  40   39  38  37  36  35  34  33  32  31  |      | 72=Ladder to 91        64=Snake to 44 |")
        print("|  21   22  23  24  25  26  27  28  29  30  |      | 80=Snake to 23         95=Snake to 75 |")
        print("|  20   19  18  17  16  15  14  13  12  11  |      | 38=Ladder to 71        98=Snake to 79 |")
        print("|  1    2   3   4   5   6   7   8   9   10  |      | 100=END                              |")
        print("**WINNER WILL BE THE ONE WHO GETS TO 100 FIRST.THAT IS WITH MINIMUM NUMBER OF DICE THROWS..\n")
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    # Start the game loop
    while position < 100:
      input("\nPress any key to roll the dice\n")
      print("Player rolled a dice.")
      dice = random.randint(1, 6)
      print(f"The number is {dice}.")
      position += dice

      # If you roll a 6, you get another chance to roll
      if dice == 6 and position<100:
        newposition = check_position(position)
        position = newposition
        print("\n6 came up! You get another chance to roll.")
        continue
      if dice<=6 and position>100:
        position=position-dice
        print(f"You are still at {dice}.")
        continue
      # Check the position for ladders or snakes
      print(f"\nYou are landed on {position}.")
      count+=1
      newposition = check_position(position)
      position = newposition

    print(f"\nCONGRATULATIONS, you have won the game in {count} throws!")

  else:
    print("\nOK, Go to Another Game!")

if __name__ == "__main__":
  main()
```

Key Points to Note:

Input/Output: In C, there are scanf and printf for input and output, respectively. In Python, input() is used for getting user input and print() for displaying messages.

Random Dice Rolls: In C, rand() is used to generate random numbers, while in Python, random.randint(1, 6) is the equivalent.

Game Loop: The game loop continues while the player's position is less than 100. The player's position updates based on the dice roll and the ladder or snake positions.

Ladders and Snakes: The check_position() function in Python handles updating the player's position if they land on a ladder or snake.

Throw Counting: The game tracks the number of dice throws. However, rolls of 6 do not count towards the throw count. Additionally, if a player's position exceeds 100 after a dice roll, that throw is not counted. But, if you are on 94 then you rolled 6,then that throw is counted.

How to Play:

The player is prompted whether they want to play.

The player rolls a dice by pressing any key. The dice roll determines the number of steps they move forward.

If they land on a snake, they move backwards; if they land on a ladder, they move forwards.

The goal is to reach position 100 as quickly as possible.